

Meateka: Machine Learning Report for Advisor

Simple Summary of Data Cleaning, Model Choice, and Findings • Author: Samol Tylundy • CADT • September 2026

1. What I Learned from EDA (Data Cleaning & Understanding)

When exploring the dataset, I found three important problems and fixed them before training the model:

Problem Found	What Happened in Data	How I Fixed It
Broken Records	248 posts had more likes than views (e.g. 18,977 likes on 8,088 views). In real social media, this is impossible because you cannot like a post without viewing it first.	I removed these 248 broken rows. The clean dataset now has 2,777 valid posts .
Crazy High Numbers	Some viral posts had engagement rates up to 3,510%. If we keep these huge numbers, they pull the averages and ruin the clustering.	I capped the engagement rate at 200.0% . This covers 98.7% of normal posts and keeps the numbers fair and balanced.
No Cheating (Target Leakage)	Likes, views, comments, and shares only happen AFTER a post is published. A creator writing a new draft does not know these future numbers yet.	Our clustering model only uses 13 pre-posting details . We only use likes and views at the end to see how each group performed in the past.
Settings vs. Virality	Post details (like posting time or hashtag count) only give the post format. They do not guarantee virality ($\$R^2 < 0.05\$$).	Clustering groups posts by creator style and format , rather than promising that a post will go viral.

2. Why Did I Choose Hierarchical Clustering Over K-Means?

I tested both K-Means and Hierarchical Clustering (Ward method) with 2 to 6 clusters (K=2 to K=6) using the 13 pre-posting features. Here are the actual test scores:

Algorithm	K	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑	Cluster Sizes
K-Means	2	0.1544	2.2779	271.23	[422, 2355]
K-Means	3	0.1028	2.6820	259.07	[422, 721, 1634]
K-Means	4	0.1082	2.4830	253.96	[530, 1233, 422, 592]
Hierarchical (Ward)	2	0.1544	2.2779	271.23	[2355, 422]
Hierarchical (Ward) *	3	0.1227	2.4807	272.37	[1756, 422, 599] (Selected)
Hierarchical (Ward)	4	0.0961	2.6232	243.30	[1248, 422, 599, 508]
Hierarchical (Ward)	5	0.0739	2.6841	210.85	[914, 422, 599, 508, 334]

1. Hierarchical Wins at K=3: When we use 3 clusters, Hierarchical Clustering beat K-Means on all three metrics: 19% higher Silhouette Score (0.1227 vs 0.1028), lower Davies-Bouldin index (2.4807 vs 2.6820 meaning less cluster overlap), and higher Calinski-Harabasz score (272.37 vs 259.07 meaning better separated clusters).

2. Why 3 Clusters Instead of 2? K=2 has a slightly higher silhouette score (0.1544), but it only separates huge celebrity accounts (422) from all normal creators (2,355). That is not useful advice for creators. K=3 gives the highest Calinski-Harabasz score (272.37) and creates three clear, useful tiers: **Low, Medium, and High**.

3. Ward Linkage Makes Clean Clusters: Ward's method groups posts by keeping the variance inside each cluster small. This stops clusters from becoming long and messy, creating neat and balanced groups.

3. What Are My Findings? (Cluster Interpretation)

The model found 3 natural groups using only the 13 pre-posting features. Afterward, we checked their historical average engagement rate:

Cluster ID	Tier	Count (N)	Mean Eng.	Median Eng.	What This Group Looks Like
Cluster 0	Low	1,756 (63.2%)	11.20%	3.10%	Standard posts: Photos and videos on Instagram and Facebook with normal caption lengths.
Cluster 1	Medium	422 (15.2%)	11.36%	2.58%	Big creator posts: Average 252,000 followers, lots of hashtags (14.7), fashion and lifestyle carousels.
Cluster 2	High	599 (21.6%)	13.42%	3.18%	High-discovery video posts: TikTok and Instagram videos/reels, tech and entertainment, posted in evening hours.

Honest Note on Low vs. Medium: Cluster 0 (11.20%) and Cluster 1 (11.36%) have almost identical engagement rates. Statistical tests (ANOVA $p=0.27$, t-test $p=0.92$) show they are not significantly different. This proves the model groups posts by creator style and format, not by guaranteed virality. Low/Medium/High is a helpful benchmark based on history, not a guarantee.

How New Drafts Are Predicted: When a user inputs a new draft, our model compares it to the 3 cluster centers using distance. It takes less than 5 milliseconds. The confidence score (e.g. 45%) shows how close the draft is to that cluster center, not a percentage probability of going viral.